Midterm 2

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```r
library(tidyverse)
library(lubridate)
library(ggplot2)
member <- read.csv("https://collinn.github.io/data/member.csv")
claims <- read.csv("https://collinn.github.io/data/claims.csv")
```

Filtering the Data

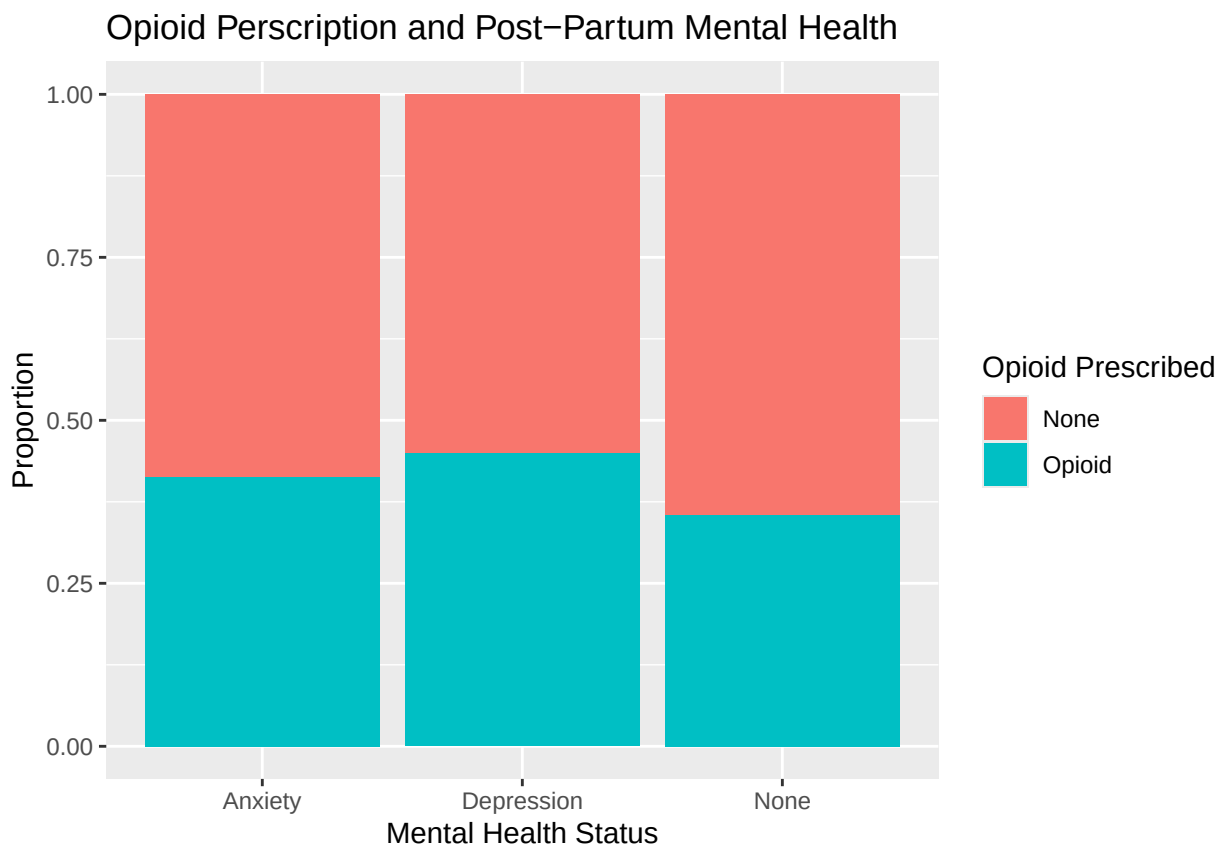
```r
# Merge both the datasets together
all_data <- left_join(claims, member, by = "id")

# Calculating each observations age at delivery
inAgeRange <- all_data %>%
  mutate(ageAtTimeOfBirth = floor(time_length(interval(ymd(DoB), ymd(date)), "years"))) %>%
  filter(ageAtTimeOfBirth >= 18 & ageAtTimeOfBirth < 40.1966)

# Insurance Coverage (given from instructions)
happen_after_n <- function(y, m, d, a, b) {
  m <- m + 12 * (y == 2018)
  event <- month(d) + 12 * (year(d) == 2018)
  if (max(event) - min(m) >= a) {
  if (max(m) - max(event) >= b) {
  if (max(event) %in% m) return(TRUE)}}
  return(FALSE)}

# Filtering for observations that meet the insurance of 13 months
filtered_data <- inAgeRange %>%
  filter(ICD %in% c("O80", "O82")) %>%
  group_by(id) %>%
  summarise(insured = happen_after_n(year, month, date, 9, 3), .groups = "drop") %>%
  filter(insured == TRUE)

# New dataset for delivery type
delivery_type <- claims %>%
  filter(ICD %in% c("O80", "O82")) %>%
  mutate(delivery_type = ifelse(ICD == "O82", "Cesarean", "Vaginal")) %>%
  group_by(id) %>%
  summarise(delivery_type = first(delivery_type), .groups = "drop")
```

```r
# New dataset for mental health diagnosis
mental_health_diagnosis <- claims %>%
  filter(ICD %in% c("F41.8", "F32.3")) %>%
  mutate(Mental_Health_Diagnosis = case_when(
    ICD == "F41.8" ~ "Anxiety",
    ICD == "F32.3" ~ "Depression")) %>%
  group_by(id) %>%
  summarise(Mental_Health_Diagnosis = first(Mental_Health_Diagnosis), .groups = "drop")

# New dataset for migraine status
migraine_status <- claims %>%
  filter(ICD == "G43.4") %>%
  mutate(Migraine_status = "Migraine") %>%
  distinct(id, Migraine_status)

# New dataset for opioid prescriptions
opioid <- claims %>%
  filter(ICD == "J0745") %>%
  mutate(Opioid_status = "Opioid") %>%
  distinct(id, Opioid_status)

# Merging all covariate datasets together for final dataset
final_set <- inAgeRange %>%
  filter(id %in% filtered_data$id) %>%
  distinct(id, .keep_all = TRUE) %>%
  left_join(delivery_type, by = "id") %>%
  left_join(mental_health_diagnosis, by = "id") %>%
  left_join(migraine_status, by = "id") %>%
  left_join(opioid, by = "id") %>%
  select(id, DoB, ageAtTimeOfBirth, PlanType, delivery_type,
         Mental_Health_Diagnosis, Migraine_status, Opioid_status)

# replace NA with "None"
final_set <- final_set %>%
  mutate(across(c(Mental_Health_Diagnosis, Migraine_status, Opioid_status),
                ~ if_else(is.na(.), "None", .)))
# Checking to see if 4246 rows printed
nrow(final_set)
```

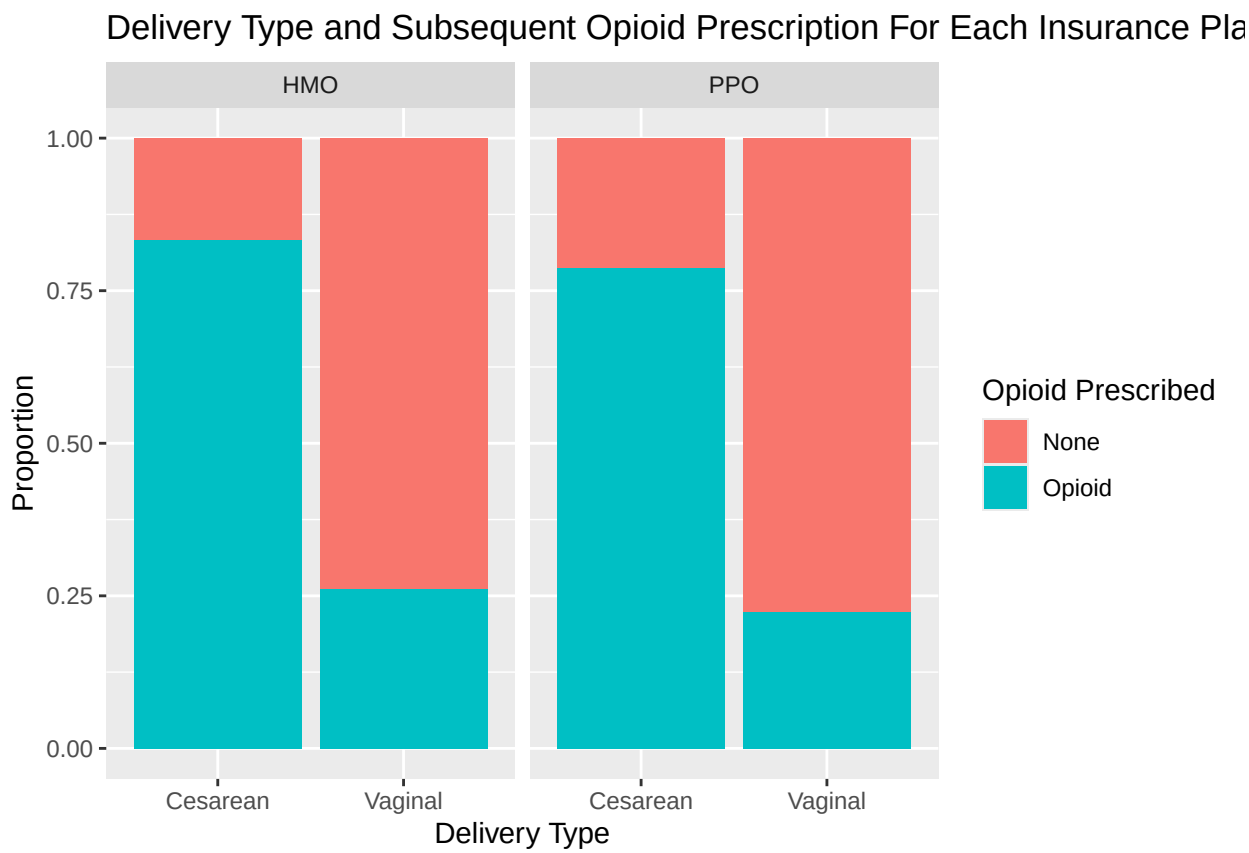
```
## [1] 4246
```

Creating Exploratory Plots

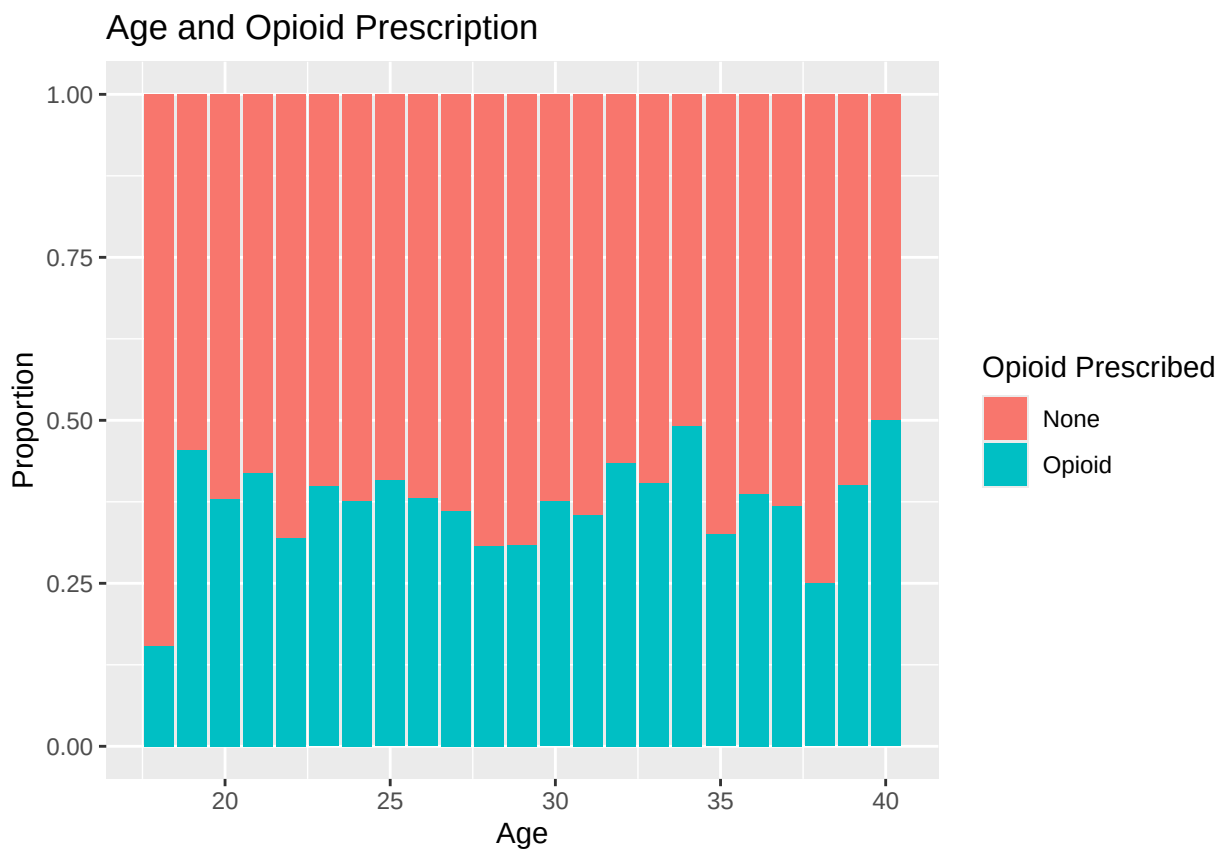
```r
#Plot 1: Opioid Perscription and Post-Partum Mental Health
ggplot(final_set, aes(x = Mental_Health_Diagnosis, fill = factor(Opioid_status))) +
  geom_bar(position = "fill")+
 labs(title="Opioid Perscription and Post-Partum Mental Health",
       x = "Mental Health Status",
       y = "Proportion",
       fill = "Opioid Prescribed")
```




```
#Plot 2: Delivery Type and Subsequent Opioid Prescription
ggplot(final_set, aes(x = delivery_type, fill = factor(Opioid_status))) +
  geom_bar(position = "fill")+
 labs(title="Delivery Type and Subsequent Opioid Prescription For Each Insurance Plan",
       x = "Delivery Type",
       y = "Proportion",
       fill = "Opioid Prescribed")+
  facet_wrap(~PlanType)
```




```
#Plot 3: Age and Opioid Prescription By Insurance Type
ggplot(final_set, aes(x=ageAtTimeOfBirth, fill = factor(Opioid_status))) +
  geom_bar(position = "fill")+
  labs(title="Age and Opioid Prescription",
       x = "Age",
       y = "Proportion",
       fill = "Opioid Prescribed")
```



Creating indicator dataset for modelling purposes

```
# C-section is stored as 1, vaginal birth is stored as 0
model_delivery_type <- claims %>%
  filter(ICD %in% c("O80", "O82")) %>%
  mutate(delivery_type = ifelse(ICD == "O82", 1, 0)) %>%
  group_by(id) %>%
  summarise(delivery_type = first(delivery_type), .groups = "drop")

# Anxiety is stored as 2, Depression is stored as 1
model_mental_health_diagnosis <- claims %>%
  filter(ICD %in% c("F41.8", "F32.3")) %>%
  mutate(Mental_Health_Diagnosis = 1)%>%
  group_by(id) %>%
  summarise(Mental_Health_Diagnosis = first(Mental_Health_Diagnosis), .groups = "drop")

# New model dataset for migraine status, stored as 1
model_migraine_status <- claims %>%
  filter(ICD == "G43.4") %>%
  mutate(Migraine_status = 1) %>%
  distinct(id, Migraine_status)

# New model dataset for opioid prescriptions, stored as 1
model_opioid <- claims %>%
  filter(ICD == "J0745") %>%
  mutate(Opioid_status = 1) %>%
  distinct(id, Opioid_status)
```

```r
# Merging all covariates together for final dataset
model_final_set <- inAgeRange %>%
  filter(id %in% filtered_data$id) %>%
  distinct(id, .keep_all = TRUE) %>%
  left_join(model_delivery_type, by = "id") %>%
  left_join(model_mental_health_diagnosis, by = "id") %>%
  left_join(model_migraine_status, by = "id") %>%
  left_join(model_opioid, by = "id") %>%
  select(id, DoB, ageAtTimeOfBirth, PlanType, delivery_type,
         Mental_Health_Diagnosis, Migraine_status, Opioid_status)

# replace NA with "None"
model_final_set <- model_final_set %>%
  mutate(across(c(Mental_Health_Diagnosis, Migraine_status, Opioid_status),
                ~ if_else(is.na(.), 0, .)))
# Checking to see if 4246 rows printed
nrow(model_final_set)
```

```
## [1] 4246
```

Running the Statistical Model

```r
#Running Logistic model to quantify covariate effect on Opioid Perscription

logit_mod <- glm(Opioid_status ~ Mental_Health_Diagnosis + ageAtTimeOfBirth + delivery_type + PlanType,

summary(logit_mod)
```

```
##
## Call:
## glm(formula = Opioid_status ~ Mental_Health_Diagnosis + ageAtTimeOfBirth +
##     delivery_type + PlanType, family = binomial, data = model_final_set)
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)             -1.164146   0.282723  -4.118 3.83e-05 ***
## Mental_Health_Diagnosis  0.440919   0.091002   4.845 1.27e-06 ***
## ageAtTimeOfBirth         0.001832   0.010377   0.177  0.85986
## delivery_type            2.597175   0.089216  29.111  < 2e-16 ***
## PlanTypePPO             -0.230633   0.081039  -2.846  0.00443 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 5598.0  on 4245  degrees of freedom
## Residual deviance: 4495.1  on 4241  degrees of freedom
## AIC: 4505.1
##
## Number of Fisher Scoring iterations: 4
```

```r
#Interpreting the results

exp(0.4401512) #calculating the exponential of Mental_Health_Diagnosis value from model
```

```
## [1] 1.552942
```

```r
exp(2.5946974) #calculating the exponential of delivery_type value from model
```

```
## [1] 13.39253
```

```r
exp(-0.230633) #calculating the exponential of PlanType (Insurance) value from model
```

```
## [1] 0.7940308
```

Discussion

In working with two datasets containing information on mothers who gave birth, we wanted to determine how opioid prescription was related to other covariates in the data. After filtering the data, we decided to perform a logistic regression, comparing opioid prescription with the patient's mental health, their age, and the type of delivery. The reason we chose to perform a logistic regression over linear regression is that logistic regression is better suited to a binomial outcome, which we had in this dataset: an opioid was either prescribed to the patient, or not.

After performing the logistic regression on our data with opioid prescription as the dependent variable, we found that a mental health diagnosis was discernibly predictive in relation to opioid prescription, with a p-value of 1.27*e^-6. After calculating the exponential value associated with mental_health_diagnosis from our regression, we found that any mental health diagnosis increased the chance of an opioid prescription by approximately 55.29%, holding all other variables constant. Thus, while a mental health diagnosis has a discernable effect on opioid prescription, it cannot be labeled a significant predictor of opioid prescription. Next, we found that age, with a p-value of 0.884, is not a discernable predictor of opioid prescription. Finally, we found that delivery type, like mental health diagnosis, was a discernable predictor of opioid prescription, with a p-value near 0. After calculating the exponential value associated with delivery_type from our regression, we found that patients who gave birth via C-section are approximately 13.39x more likely to get an opioid prescription than those who gave vaginal births, holding all other variables constant. Thus, the birth type is a highly significant predictor for whether patients were prescribed opioids, with mothers who had a C-section being overwhelmingly more likely to be prescribed opioids compared to mothers who gave vaginal birth. Finally, we included PlanType, or the kind of insurance the mothers had, in our statistical model. We found that the PPO plan type, with a p-value of 0.00443, had a discernible effect on opioid prescription. However, in exponentiating, we find that there is little effect of having the PPO insurance plan on opioid prescription, compared to patients with the HMO insurance plan, thus insurance type, like age and mental health is not a significant predictor of opioid prescription.

Finally, for our exploratory plots, we decided to visualize the same covariates we included in our statistical model. Our three exploratory plots align with the outcome of our regression: having a C-section delivery type appeared strongly correlated with opioid prescription in the visualization we created, which aligned with the logistic regression results. We decided to facet this visualization by insurance type, which showed almost no difference between insurance types, which was also consistent with our logistic regression results. Our visualization for mental health suggested that there may exist some small correlation with opioid prescription and a mental health diagnosis, which our model confirmed. Finally, our visualization comparing the age of patients and opioid prescription showed no discernable correlation between the two covariates, which our statistical model confirms.